

From hyperspectral CHRIS to four RGBN bands

The cloud model eats **4 bands** (R, G, B, NIR). CHRIS has **18 to 62 bands** per mode. So each pseudo-band is built from the CHRIS bands that fall in it.

Two ways to build them: the plain mean the model was trained on, then SRF-weighted resampling.

METHOD 1 OF 2

Plain mean

which each CHRIS band goes to the Sentinel-2 band it overlaps the most

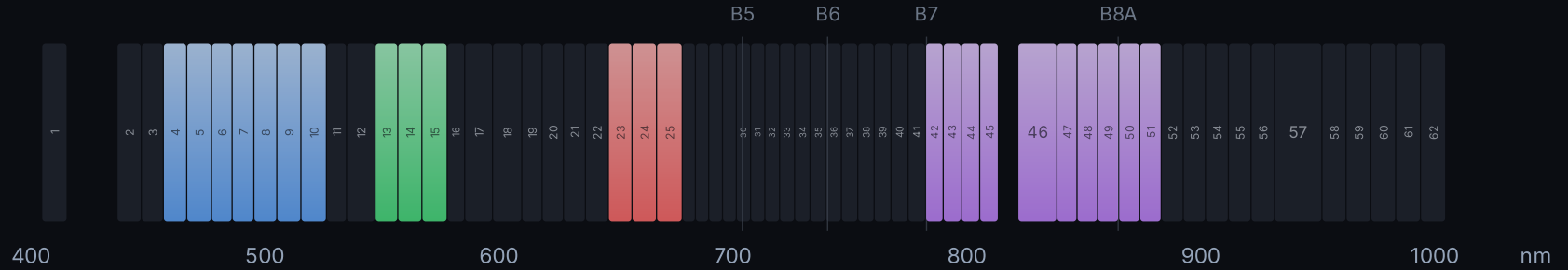
how plain **mean** of the CHRIS bands assigned to each pseudo-band

then DN/100 000 or TOA/10 000, clip to 5, and a per-tile z-score inside the network

$$\bar{L}_g = \frac{1}{N_g} \sum_{i \in g} L_i$$

This is the training mapping, the one the model actually learned.

Mode 1 : 62 bands, ~34 m



B2
459-525 nm



B3
542-578 nm

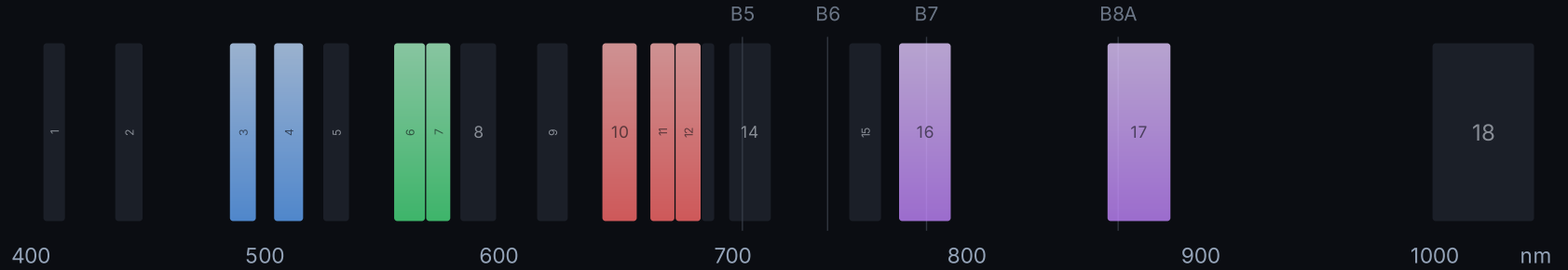


B4
649-680 nm



B8
780-886 nm

Mode 2 : 18 bands, ~17 m



B2

459-525 nm



B3

542-578 nm



B4

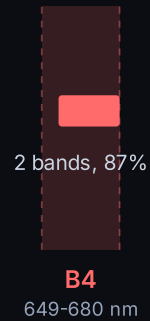
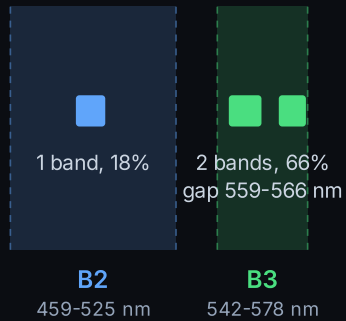
649-680 nm



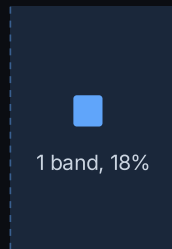
B8

780-886 nm

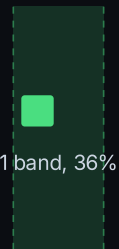
Mode 3 : 18 bands, ~13 m



Mode 4 : 18 bands, ~11 m



B2
459-525 nm



B3
542-578 nm

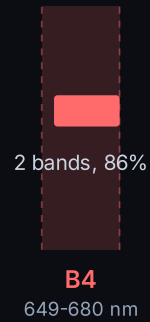
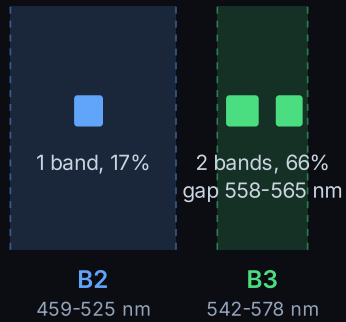
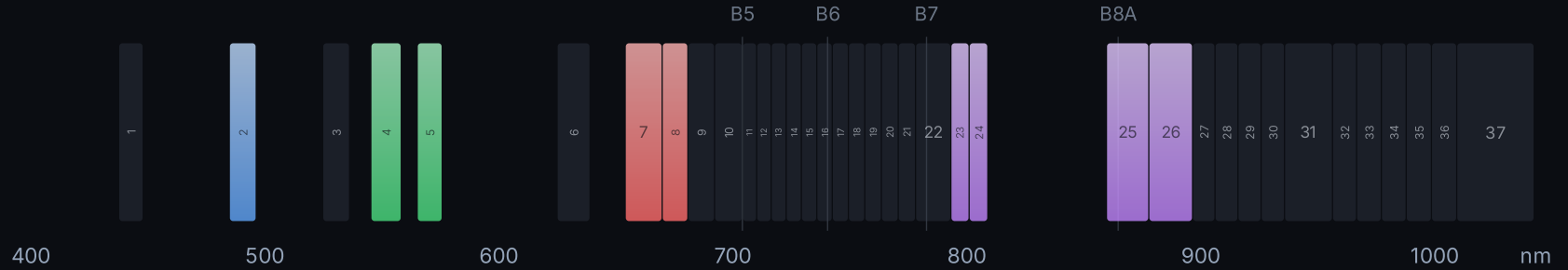


B4
649-680 nm

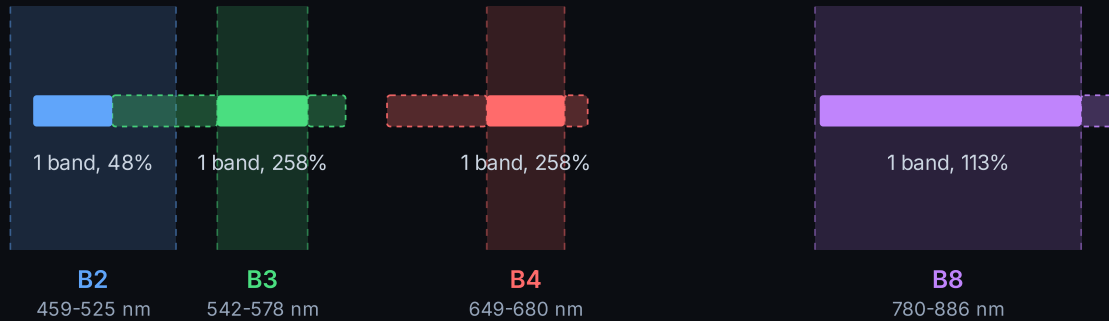


B8
780-886 nm

Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Method 1: where the bands end up

R, B4

G, B3

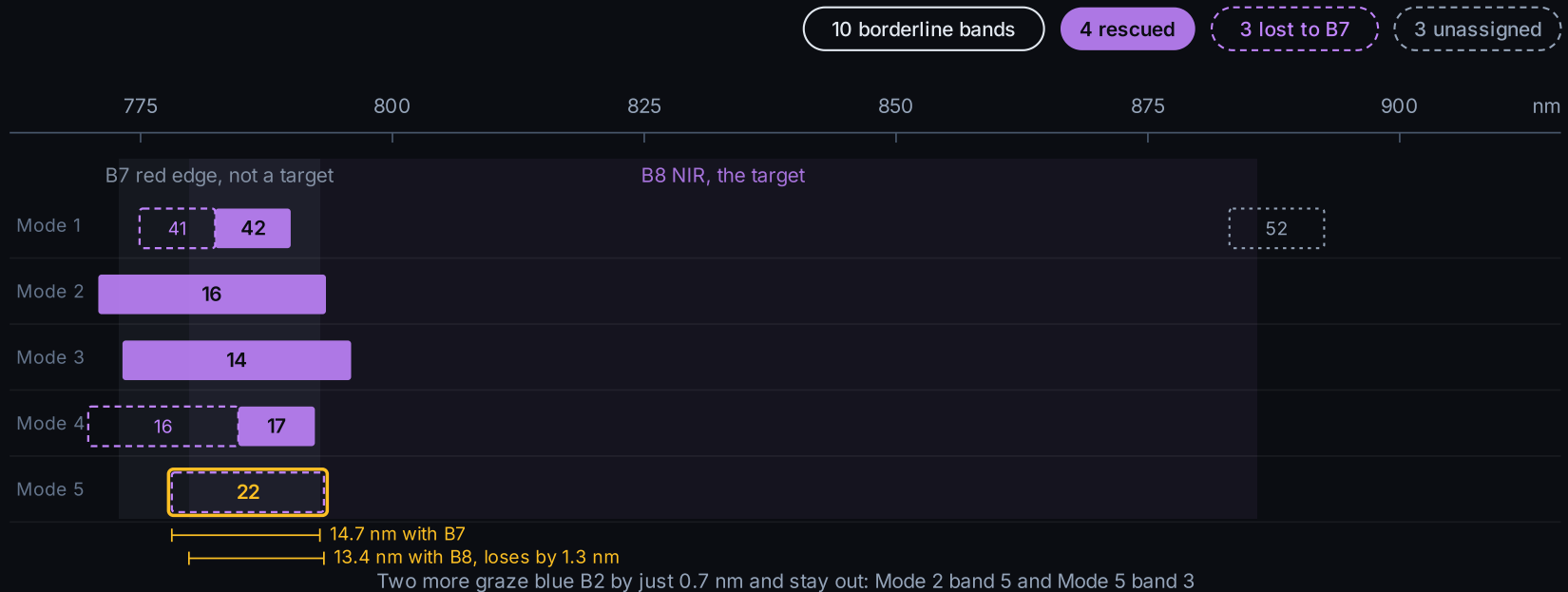
B, B2

NIR, B8

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	4-10 (7), 457-526 nm	13-15 (3), 547-578	23-25 (3), 647-678	42-51 (10), 782-883
2	3-4 (2), 485-517	6-7 (2), 555-580	10-12 (3), 644-686	16-17 (2), 771-887
3	2 (1), 486-497	4-5 (2), 546-577	7-8 (2), 656-683	14-15 (2), 773-890
4	1 (1), 484-496	2 (1), 545-558	4 (1), 664-675	17-18 (2), 785-800
5	2 (1), 485-496	4-5 (2), 545-576	7-8 (2), 654-681	23-26 (4), 793-896
20	1 (1), 469-500	2 (1), 500-593	3 (1), 609-689	4 (1), 782-901

Training mapping. Numbers are 1-based CHRIS band indices; nm is the spectral span. Stack order: R, G, B, NIR.

What the max overlap rule leaves out



Each of these bands touches the NIR target but the winner takes all rule sends it away. Four were rescued by hand for training, three were left out after losing the red edge B7 duel (including Mode 5 band 22, the heaviest NIR band the SRF sees), and three stay unassigned under the 30 percent overlap threshold.

METHOD 2 OF 2

SRF-weighted resampling

which every CHRIS band with nonzero S2A response takes part, no winner takes all

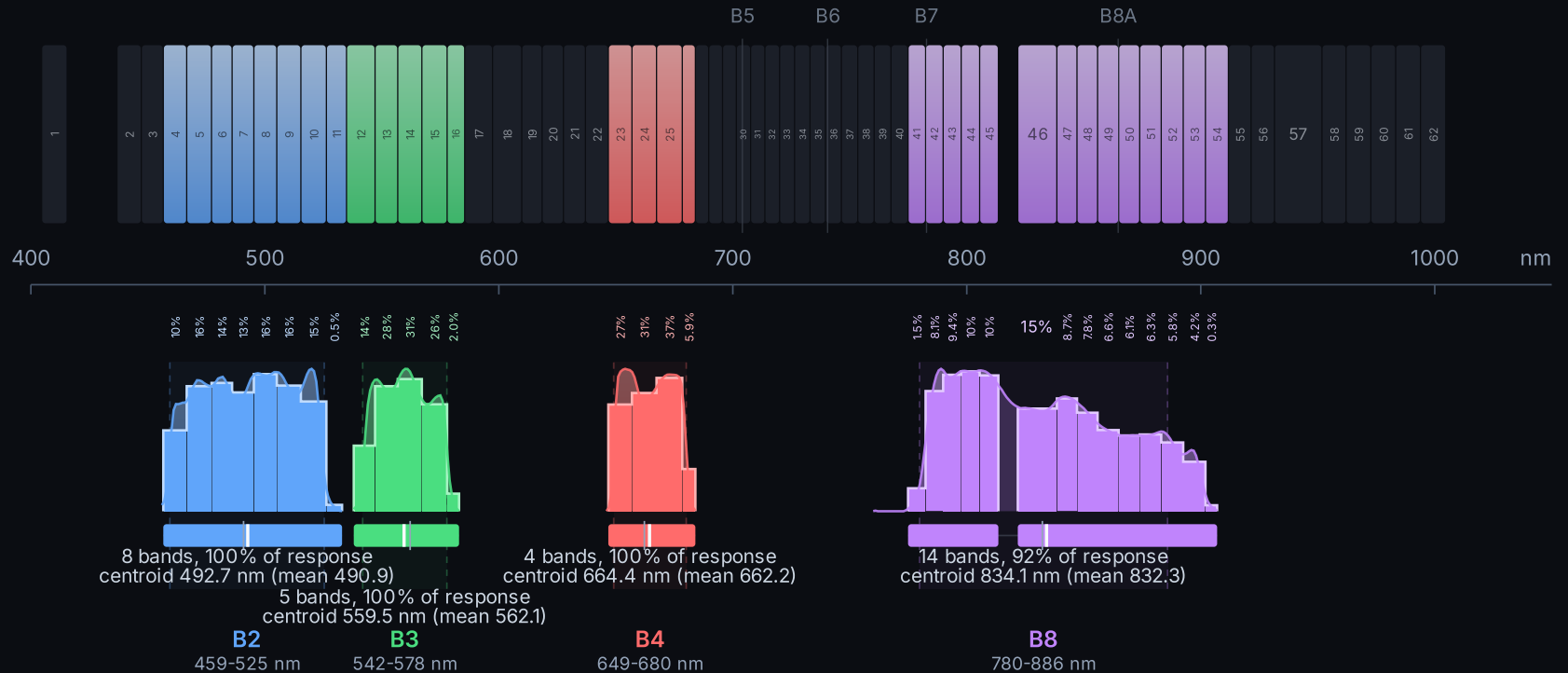
how w_i is the integral of the S2A response curve over the CHRIS band, and the pseudo-band is the weighted mean

then weights are normalized by the area CHRIS actually covers, not by the full SRF area

$$\tilde{L}_g = \frac{\sum_i w_i L_i}{\sum_i w_i}, \quad w_i = \int_{\lambda_i^{lo}}^{\lambda_i^{hi}} R_g(\lambda) d\lambda$$

Same four targets, read through the real S2A response curves instead of flat FWHM boxes.

Mode 1 : 62 bands, ~34 m



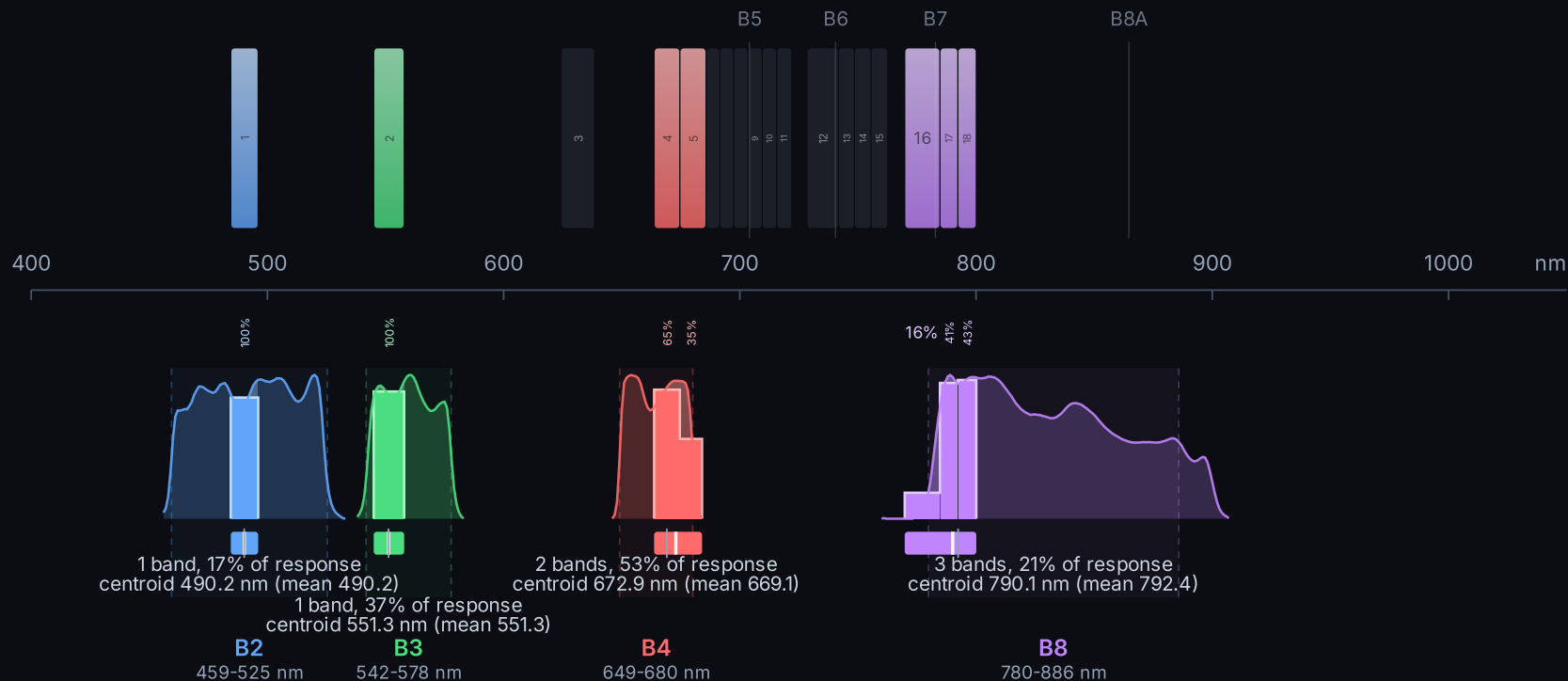
Mode 2 : 18 bands, ~17 m



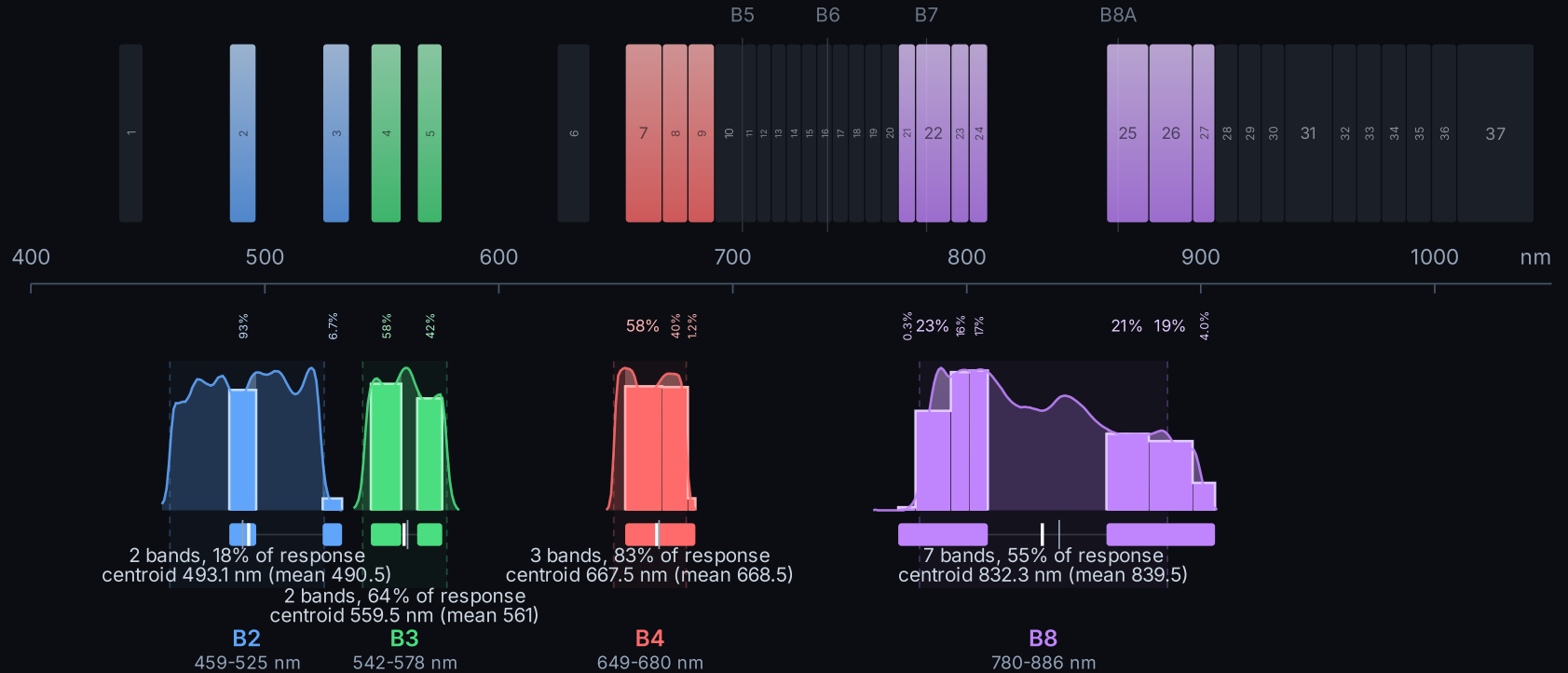
Mode 3 : 18 bands, ~13 m



Mode 4 : 18 bands, ~11 m



Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Plain mean vs SRF weights

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	7 → 8	3 → 5	3 → 4	10 → 14
2	2 → 3	2 → 2	3 → 3	2 → 2
3	1 → 2	2 → 2	2 → 2	2 → 3
4	1 → 1	1 → 1	1 → 2	2 → 3
5	1 → 2	2 → 2	2 → 3	4 → 7
20	1 → 2	1 → 1	1 → 1	1 → 1

CHRIS bands entering each pseudo-band. Left number is the plain mean set, right number the SRF set. Mode 5 band 22 comes back on its own, about a fifth of the B8 mix, and in mode 20 the SRF hands band 2 a share of the blue.